LAB 06 – Report

Analysis and report

Record which passwords were found and how quickly.

	Length	Character set	Password	Entropy	Safe?
G1	8 characters	only letters (lowercase and uppercase)	KfZaMnQr	45	Not ideal. Make it longer.
G2	12 characters	letters + numbers	t7Qp9Lm2Vr5S	71	Better, but 16+ chars is safer.
G3	16 characters	letters + numbers + special characters	Y7p!qR2#vM9$zK4@	105	Strong if unique.
G4	24 characters	anything	nP7!xQ2#vR9$zK4@tB6%wL5?	157	Very strong; best for password manager.
G5	Passphrase	4 words (separated by spaces or dashes)	orange-river-carpet-moon	134	

Activity: Setting Up MFA with Microsoft Authenticator (TOTP)

Service used: GitHub
MFA paired app: Microsoft Authenticator

Steps completed:

1. Opened GitHub security settings.
2. Chose two-factor authentication.
3. Selected authenticator app / TOTP.
4. Scanned the QR code with Microsoft Authenticator.
5. Entered the 6-digit code.
6. Saved recovery codes.
7. Logged out and tested login.

Microsoft Authenticator generates time-based one-time passwords, usually rotating every 30 seconds.

MFA questions

Why is TOTP more secure than SMS?

TOTP does not depend on phone carriers, so it is safer against SIM-swap and phone-number takeover attacks. NIST warns that SMS/PSTN authentication has risks such as SIM changes and number porting.

What happens if you lose access to Bitwarden Authenticator?

You may be locked out unless you saved recovery codes or added another MFA method. Bitwarden warns users to save recovery codes after enabling two-step login.

Do all services support TOTP?

No. Some services support only SMS, email codes, push notifications, passkeys, or hardware keys. TOTP works only when the service supports authenticator-app MFA.

Reflection and analysis

How does the security score increase as you add length?

The score increases a lot as the password gets longer. Each extra character makes the password harder to guess.

How do special characters affect the score?

Special characters usually improve the score because they add complexity. However, they do not help much if the password is still short or predictable, like Password1!.

How does a passphrase score compare to a classic password?

A good passphrase often scores better than a short classic password because it is much longer. It is also easier to remember, especially if the words are random.

Which password would you recommend for everyday use and why?

I would recommend a long random password stored in Bitwarden for normal accounts. For a password you must remember, I would use a long passphrase because it is strong and easier to type.